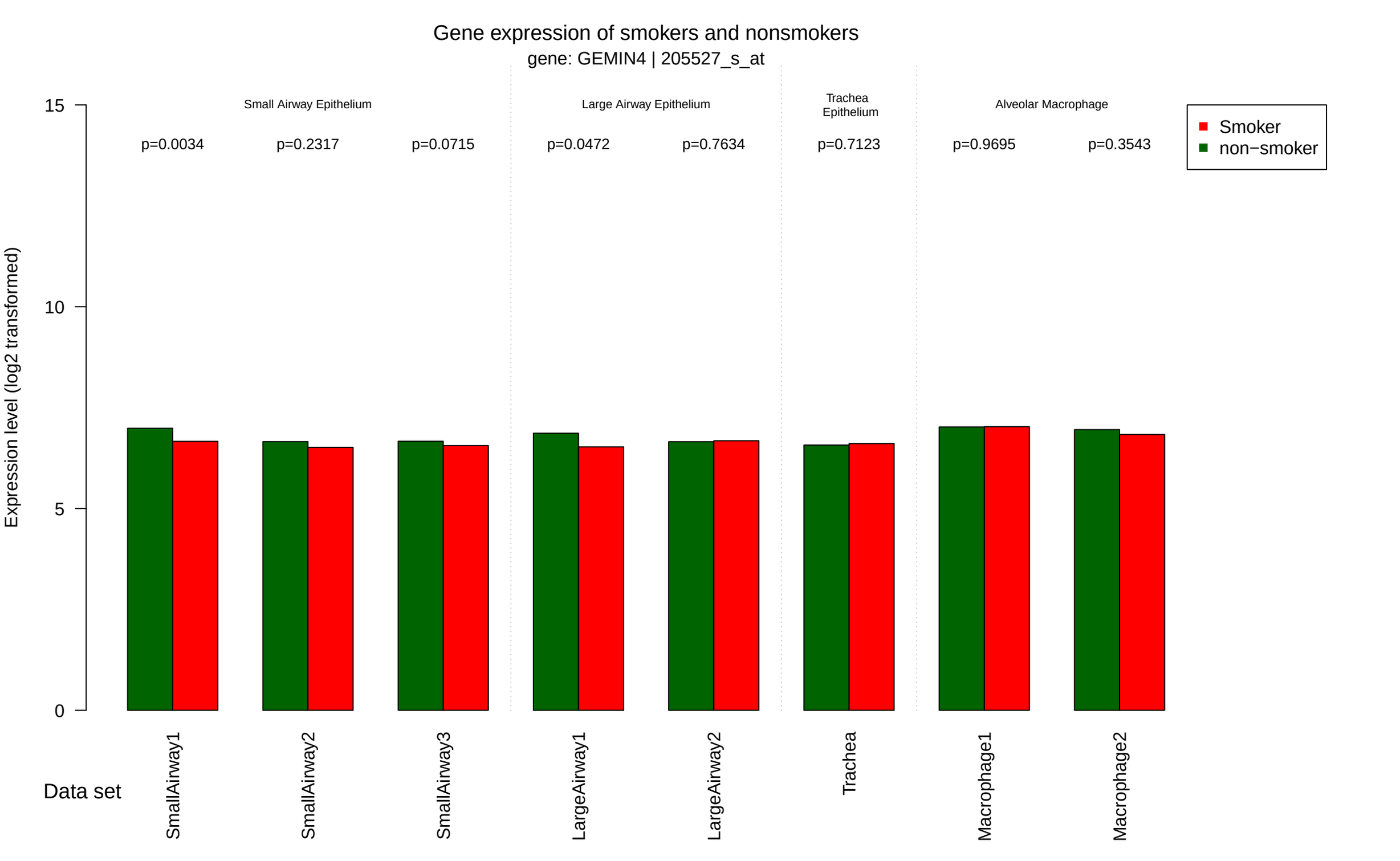
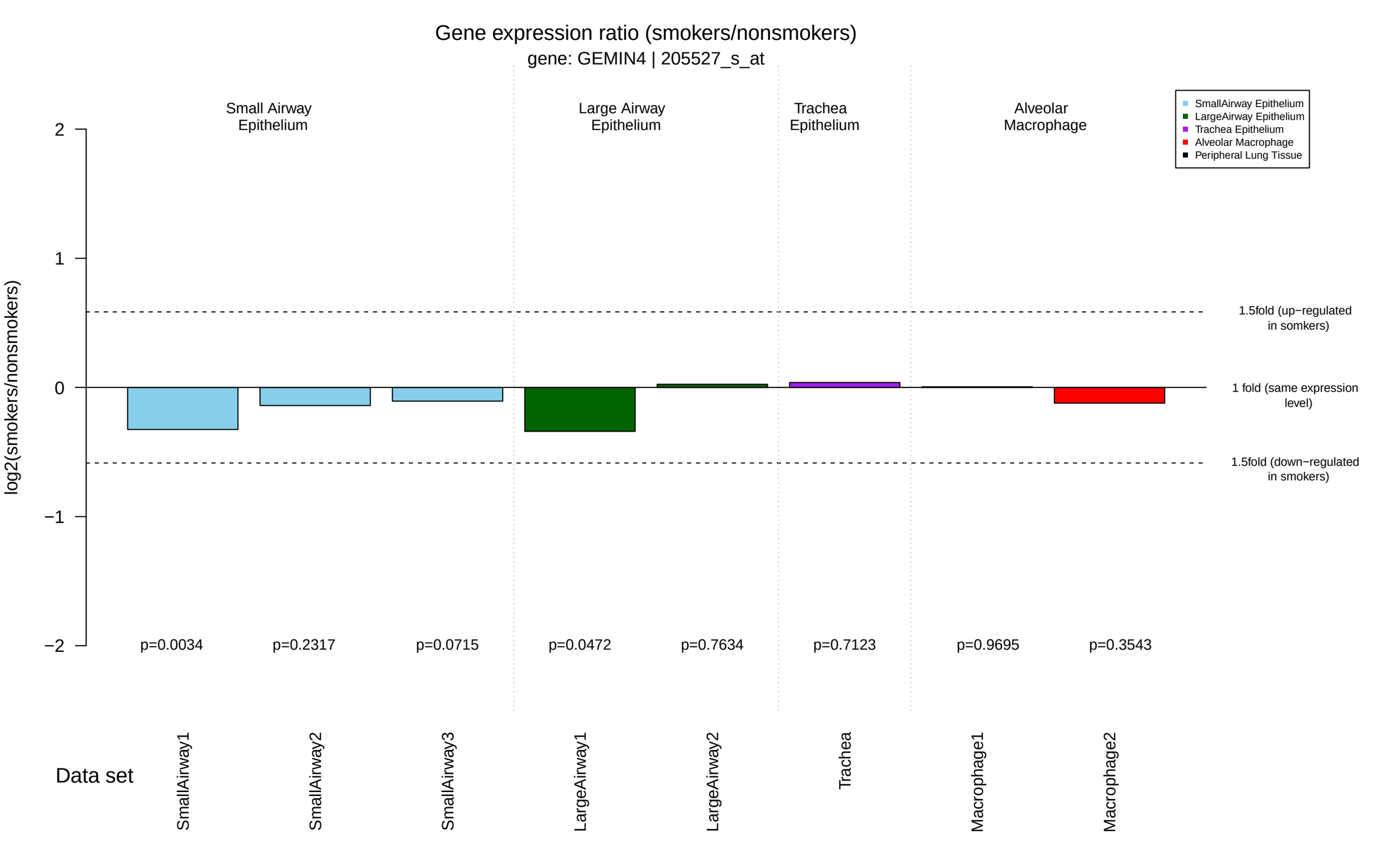
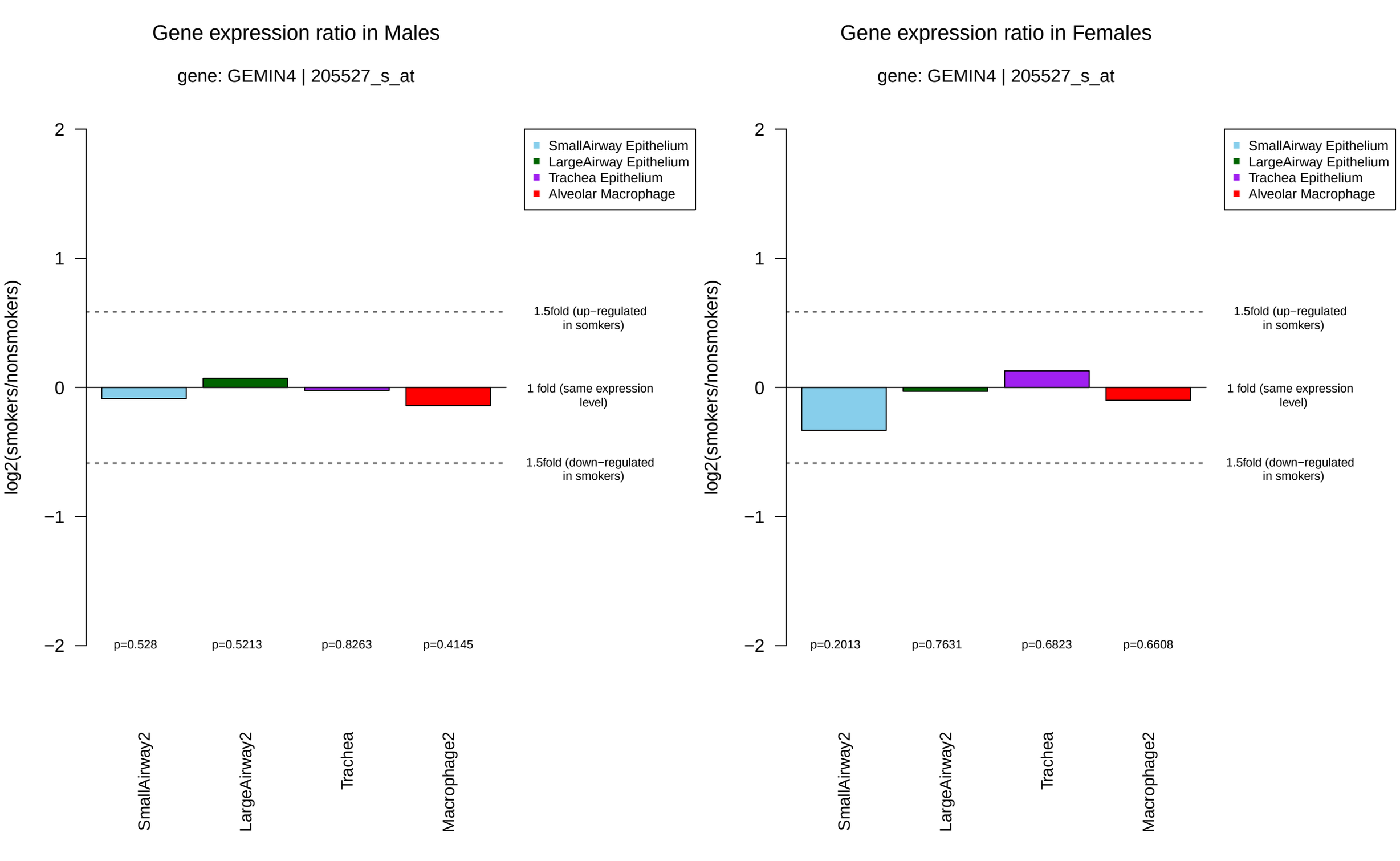
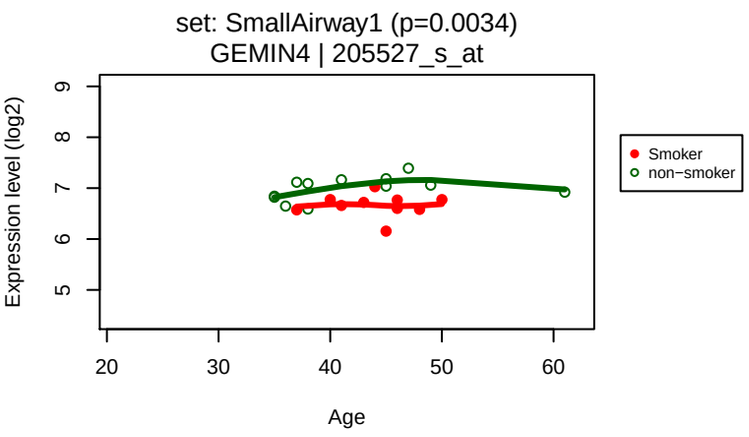
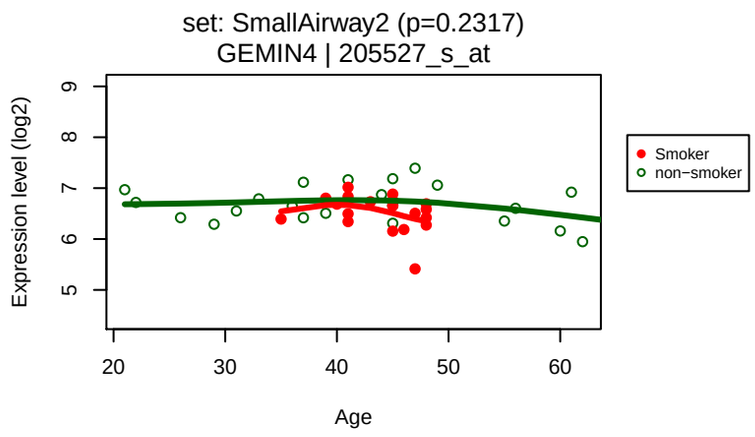
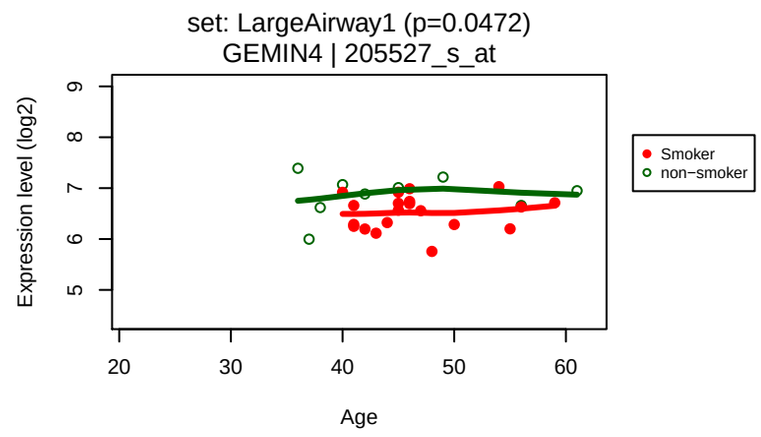

Gene expression of smokers and nonsmokers
gene: GEMIN4 | 205527_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0034
p=0.2317
p=0.0715
p=0.0472
p=0.7634
p=0.7123
p=0.9695
p=0.3543
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

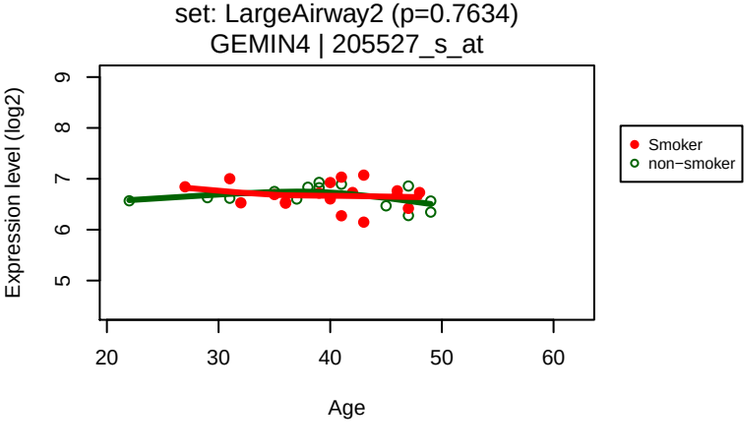

Gene expression ratio (smokers/nonsmokers)
gene: GEMIN4 | 205527_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0034
p=0.2317
p=0.0715
p=0.0472
p=0.7634
p=0.7123
p=0.9695
p=0.3543
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

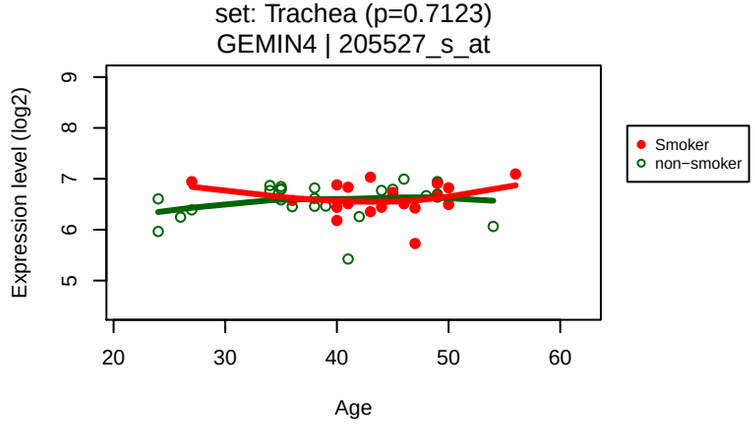

Gene expression ratio in Males
gene: GEMIN4 | 205527_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.528
p=0.5213
p=0.8263
p=0.4145
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GEMIN4 | 205527_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2013
p=0.7631
p=0.6823
p=0.6608
SmallAirway2
LargeAirway2
Trachea
Macrophage2

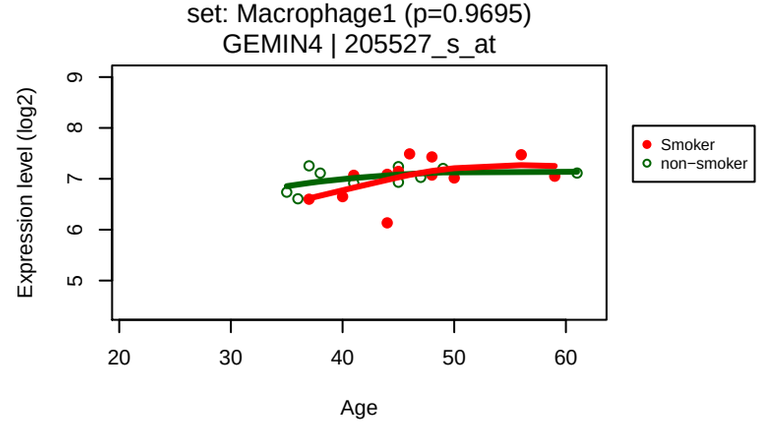

set: SmallAirway1 (p=0.0034)
GEMIN4 | 205527_s_at
Expression level (log2)
Age
Smoker
non−smoker

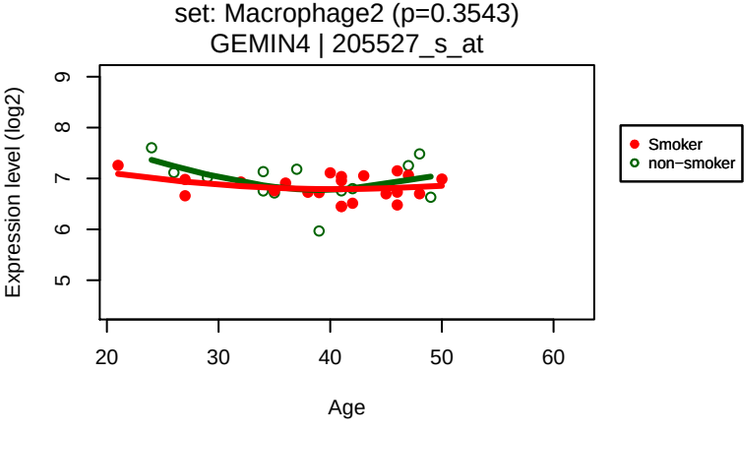

set: SmallAirway2 (p=0.2317)
GEMIN4 | 205527_s_at
Expression level (log2)
Age
Smoker
non−smoker

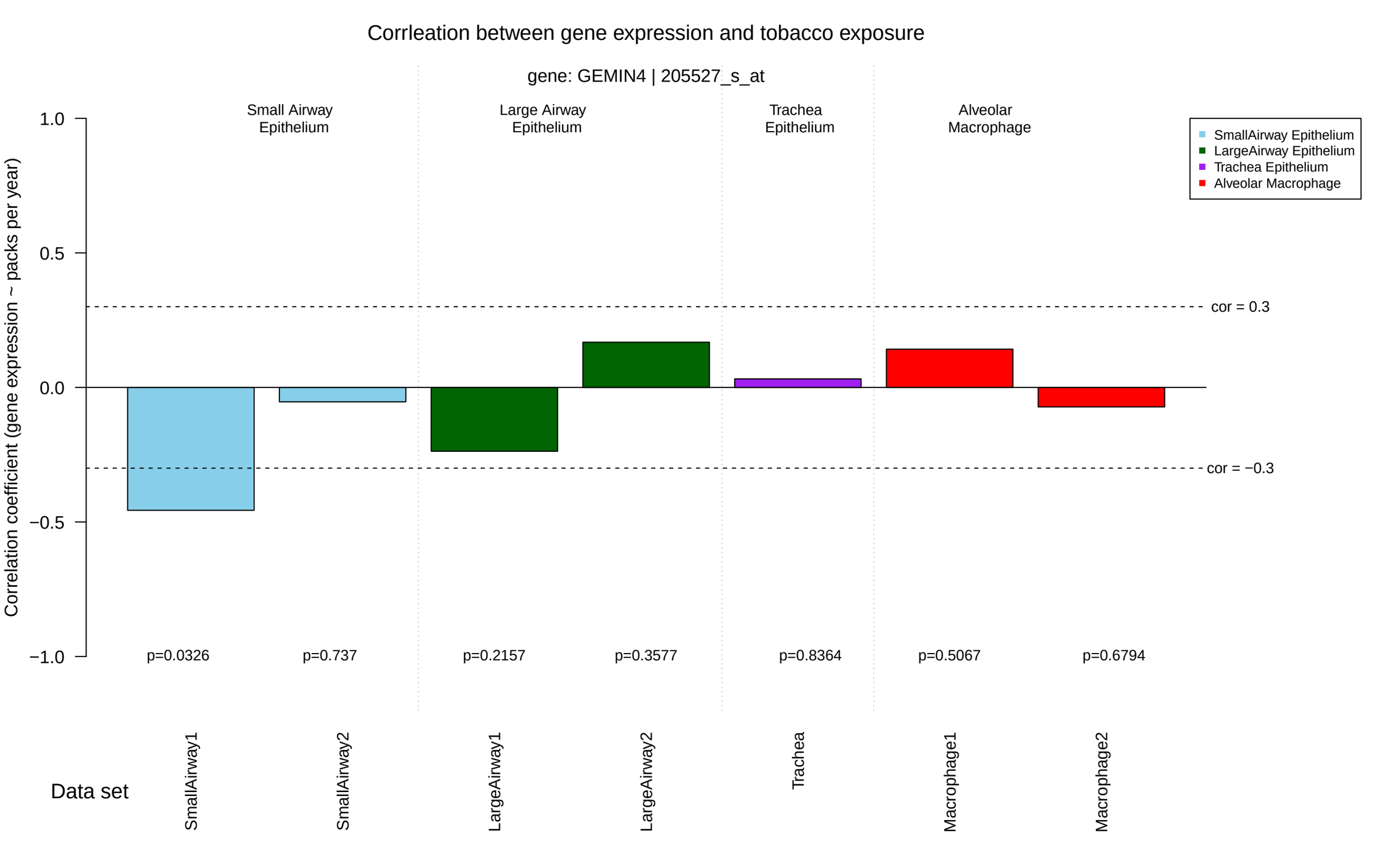

set: LargeAirway1 (p=0.0472)
GEMIN4 | 205527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7634)
GEMIN4 | 205527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7123)
GEMIN4 | 205527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9695)
GEMIN4 | 205527_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3543)
GEMIN4 | 205527_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GEMIN4 | 205527_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0326
p=0.737
p=0.2157
p=0.3577
p=0.8364
p=0.5067
p=0.6794
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2